Samir Kuinkel

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RESEARCH INTERESTS

Invasion biology, biogeography, and ecological niche modeling, with a secondary focus on urban thermal ecology and macroecological morphology of social insects. My research integrates large-scale occurrence data, species distribution modeling, computer vision, and physiological experiments to understand how species respond to environmental change.

EDUCATION

PhD in Biology Aug 2023 – Expected 2028

University of Missouri–St. Louis (UMSL), St. Louis, MO

Advisor: Dr. Sara E. Miller | GPA: 3.87/4.0

Dissertation: Thermal Ecology, Invasion Biology, and Macroecological Morphology of North American Polistes Paper Wasps.

M.Sc. AgriGenomics Oct 2018 – May 2022

Christian-Albrechts-Universität, Kiel, Germany

Final grade: 1.7 (German grading scale, where 1.0 = highest and 4.0 = passing)

Thesis: CRISPR-Cas9 gene editing of Brassica napus L. SOC1 and SPL5 homeologs

B. Tech in Biotechnology Apr 2012 – Nov 2016

Kathmandu University, Dhulikhel, Nepal

GPA: 3.3/4.0

Thesis: In vivo assessment of anti-ulcer activity of Bambusa species

PUBLICATIONS

Peer-Reviewed Journal Articles

Kuinkel S, Miller SE. (2026). Niche dynamics and climatic novelty drive the invasion success of the European paper wasp across North America. *Journal of Biogeography*. Accepted. https://doi.org/10.1111/jbi.70267

Behrends G, Hagan T, Kuinkel S, Miller SE. (2025). Genomic architecture in social insects is more strongly associated with phylogeny than social behavior. *Annals of the Entomological Society of America*, 118(1), 59–72.

Upreti A, Byanju B, Pokhrel B, Kuinkel S, Bhusal A, Thapa N, Suwal MM, et al. (2016). Bambusa balcooa Roxb.: A novel remedy for peptic ulcer. *Kathmandu University Journal of Science, Engineering and Technology*, 12(2).

Manuscripts in Preparation

Kuinkel S, Miller SE. Urban heat island effects on thermal safety margins of three sympatric *Polistes* species in St. Louis, MO. (Target submission: late 2026)

Kuinkel S, Miller SE. Macroecological patterns of body coloration in North American *Polistes* paper wasps using a computer-vision-based morphometric pipeline. (In preparation)

RESEARCH EXPERIENCE

Graduate Research Assistant Aug 2023 – Present

Social Insect Diversity Lab, University of Missouri–St. Louis

Advisor: Dr. Sara E. Miller

Chapter 1: Niche Dynamics and Invasion Biology

- Compiled and cleaned 82,058 occurrence records for 16 *Polistes* species from GBIF and iNaturalist.
- Conducted comparative niche overlap analyses across all 16 North American *Polistes* species, and native–invasive niche dynamics analyses for *P. dominula* using the COUE (centroid shift, overlap, unfilling, expansion) framework.
- Quantified climatic novelty across the invaded range using Multivariate Environmental Similarity Surface (MESS) analyses and identified anthropogenic and climatic drivers of niche expansion.

Chapter 2: Urban Thermal Biology

- Deployed temperature loggers at 60 *Polistes* nest sites across the St. Louis urban heat gradient (2024–2025).
- Conducted thermal tolerance experiments (CTmax/CTmin) on 385 individuals across three acclimation temperatures.
- Performed landscape analyses using Google Earth Engine and optimized RNA extraction protocols for downstream RNA-seq.
- Built STL Wasp Watch (stlwaspwatch.github.io), a community-science platform for St. Louis residents to report wasp nest sightings.

Chapter 3: Macroecological Morphology

- Developed a four-stage computer-vision pipeline (color calibration, object detection, instance segmentation, and color extraction) to quantify body coloration from museum specimens.

Master's Researcher Jul 2021 – May 2022

Christian-Albrechts-Universität, Kiel, Germany

- Designed CRISPR-Cas9 guide RNAs targeting *SOC1* and *SPL5* homeologs in *Brassica napus*.
- Performed *Agrobacterium tumefaciens*-mediated plant transformation and molecular validation of gene edits.

CONFERENCE PRESENTATIONS

Kuinkel S, Miller SE. (2026). How close to the edge? Thermal risk in three urban paper wasps. Poster, Harris Center for Conservation Biology Gala, UMSL, St. Louis, MO.

Kuinkel S, Miller SE. (2026). The dual strategy of a super-invader: niche conservatism and expansion in the European paper wasp (*Polistes dominula*). Poster, UMSL Graduate Research Fair, St. Louis, MO.

Kuinkel S, Miller SE. (2025). Thermal niche differentiation in native and invasive paper wasps. Oral presentation, Midwest Ecology and Evolution Conference (MEEC).

Kuinkel S, Miller SE. (2025). The dual strategy of a super-invader: niche conservatism and expansion in the European paper wasp (*Polistes dominula*). Oral presentation, St. Louis Ecology, Evolution, and Conservation Symposium.

PEER REVIEW

Co-reviewed one manuscript under the mentorship of Dr. Sara E. Miller (2024–2025). Actively building independent peer-review service through journal reviewer registration.

TEACHING EXPERIENCE

Teaching Assistant — Genetics Laboratory (BIOL 2013) 2024–2025

University of Missouri–St. Louis (2 semesters)

- Led laboratory sections on Mendelian genetics and molecular techniques (PCR, transformation, restriction digestion).

Teaching Assistant — Introductory Biology (BIOL 1821) 2023–2024

University of Missouri–St. Louis (2 semesters)

- Assisted in teaching organismal biology, ecology, and evolution to first-year undergraduate students.

MENTORSHIP AND OUTREACH

Undergraduate Research Mentor 2024–2025

Social Insect Diversity Lab, UMSL

- Mentored three undergraduate researchers: two on thermal physiology experiments (CTmax/CTmin) and one on RNA extraction and cDNA synthesis protocol optimization.

CLIMB Program Mentor 2024

University of Missouri–St. Louis

- Mentored a St. Louis high school student on field ecology and laboratory techniques; mentee independently conducted thermal physiology experiments on bees.

Community Science Coordinator — STL Wasp Watch 2024 – Present

stlwaspwatch.github.io

- Built and maintain a community-science platform for St. Louis residents to log wasp nest sightings and track local species distributions.
- Maintain a lab field-observation database for tracking nest-site data across seasons.

Volunteer — Harris Center Earth Day 365 2024

Forest Park, St. Louis, MO

- Engaged St. Louis families and children with insect museum specimens at a community outreach booth.

AWARDS AND FELLOWSHIPS

Harris Center International Student Award — Harris Center for Conservation Biology, UMSL ($2,000), 2024.

John Denver Scholarship — University of Missouri–St. Louis ($2,000), 2024.

SKILLS

Computational and Data Analysis

R (statistical modeling, ggplot2, mixed models, ecospat, CoordinateCleaner, spThin), Python (intermediate), Google Earth Engine, GIS, species distribution modeling, WorldClim bioclimatic data processing, GBIF/iNaturalist data cleaning, bioinformatics pipelines, computer vision (object detection, instance segmentation), data visualization.

Field and Laboratory

CTmax/CTmin thermal tolerance assays, RNA/DNA extraction, PCR, restriction digestion, molecular cloning, RT-qPCR, RNA-seq library preparation, CRISPR-Cas9 gene editing, Agrobacterium-mediated plant transformation.

Languages

Nepali (native), English (fluent), German (intermediate).